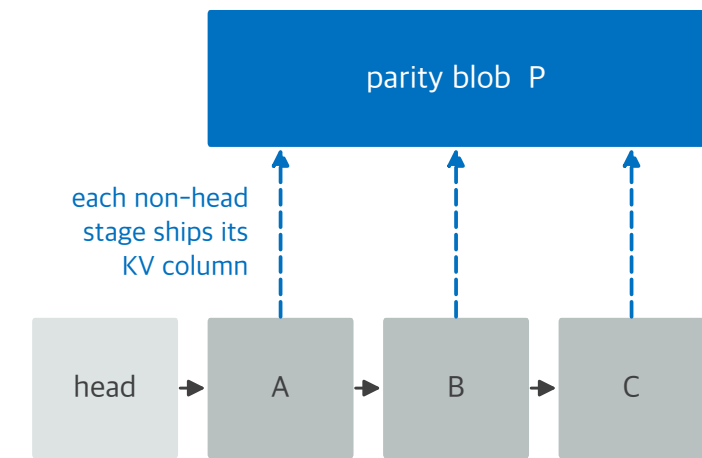


① Steady state

one blob, not one copy per stage

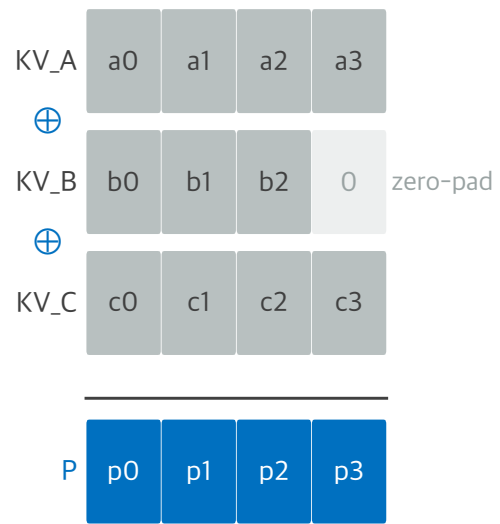


head is coordinator-sourced —
not in the parity group

$$P = KV_A \oplus KV_B \oplus KV_C$$

② Byte-wise XOR

raw bytes — works for any dtype

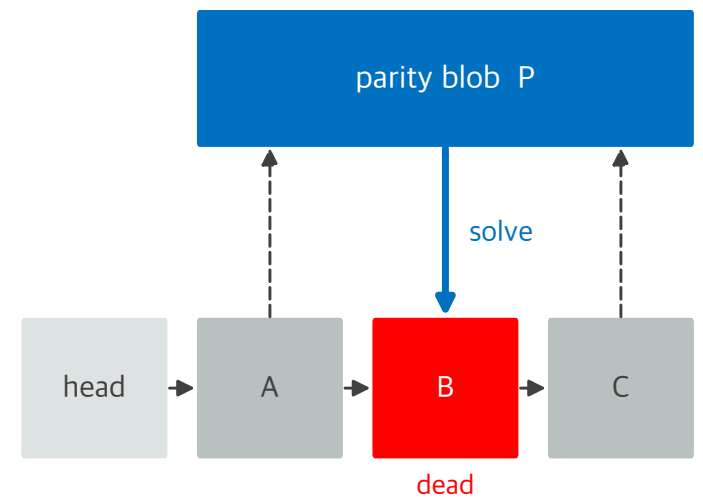


one column per position, per stage

③ Recovery

zero forward passes

survivors give back what they still hold



$$KV_B = KV_A \oplus KV_C \oplus P$$